

D552

# Data Analytics for Accountants I

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

# Items

1.

The controller of a regional grocery chain with 42 stores asks the analytics team to build a model that estimates next quarter's inventory shrinkage at each store based on past patterns. The team uses the company's cloud BI platform. Which type of analytics is this?

- A. Diagnostic.
- B. Descriptive.
- C. Prescriptive.
- D. Predictive.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2. [TRANSFER — reversed stem / different variable]

An analysis recommends the order quantity for each store that minimizes expected shrinkage cost. Which type of analytics is this?

- A. Predictive.
- B. Diagnostic.
- C. Prescriptive.
- D. Descriptive.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

The CFO of a software company with a newly built data lake wants to start by pulling every table into a model to see what patterns appear. Under the IMPACT cycle, what should come first?

- A. Perform a test plan on all available tables.
- B. Identify the business question the analysis must answer.
- C. Master the data by profiling every table.
- D. Expand the data lake so more tables are available.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

An accountant moves 18 months of expense reports from a legacy travel system into the new data warehouse. Dates in the old system are stored as text in three formats, and some employee IDs have leading spaces. The migration budget is \$40,000. In which ETL step should these problems be fixed?

- A. Load.
- B. Transform.
- C. Extract.
- D. No step, because the warehouse always fixes formats automatically.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5. [TRANSFER — reversed stem / different variable]

Clean, standardized data is written into the target data warehouse tables. Which ETL step is this?

- A. Load.
- B. Extract.
- C. Data mining.
- D. Transform.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6.

After extracting the year's 52,318 sales invoices from the ERP, an analyst loads them into a spreadsheet with 52,318 rows, but the dollar total is \$1.2 million less than the general ledger's sales account. The ERP vendor is based in Germany. What should the analyst conclude?

- A. The extract is not validated; reconcile the dollar total to the ledger (for example, missing credit memos or truncated amounts) before analysis.
- B. The data are complete, because the row counts match.
- C. Proceed; a \$1.2 million difference is always immaterial in a big data set.
- D. The ledger must be wrong, because invoices are the source records.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7.

An audit team at a national bank wants to analyze two sources: a table of loan balances with fixed columns, and thousands of scanned loan-officer notes and emails. The bank has 900 branches. Which classification is correct?

- A. The loan table is structured; the notes and emails are unstructured.
- B. Both are unstructured, because the volume is too large for a spreadsheet.
- C. Both are structured, because both are stored digitally.
- D. The notes are structured and the table is unstructured.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8.

An accounts payable team has three years of invoices, each already labeled by auditors as error or clean. They want a model that labels new invoices the same way. The invoices were scanned by an outside vendor. Which technique fits?

- A. Profiling the average invoice amount.
- B. Clustering.
- C. Classification.
- D. Regression.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9. [TRANSFER — reversed stem / different variable]

An analyst wants to estimate next month's warranty expense in dollars from sales volume. Which technique fits?

- A. Classification, since all data mining sorts records into groups.
- B. Classification.
- C. Regression.
- D. Clustering.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

10.

In a company's sales database, the Customers table has a CustomerID field, and the Invoices table also stores CustomerID on each invoice. The database runs on a server in the company's Denver office. What role does CustomerID play in the Invoices table?

- A. The primary key of the Invoices table.
- B. A foreign key linking each invoice to one customer.
- C. A composite key made of CustomerID and the server name.
- D. It must be unique in the Invoices table.

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

11. [TRANSFER — reversed stem / different variable]

Which field is the best choice for the primary key of an Employees table?

- A. Last name.
- B. Hire date.
- C. Department ID.
- D. A system-assigned employee ID number.

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

12.

A controller wants to show the board how monthly operating expenses changed over the past 24 months. The board meets quarterly at the company's new headquarters. Which chart fits best?

- A. A pie chart.
- B. A scatter plot.
- C. A table of all 24 numbers, because charts always distort data.
- D. A line chart.

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer D Concept 1

The controller of a regional grocery chain ~~with 42 stores~~ asks the analytics team to build a model that estimates next quarter's inventory shrinkage at each store based on past patterns. ~~The team uses the company's cloud BI platform.~~ Which type of analytics is this?

**Target word:** analytics type **Decides:** Estimating a future value.

|          |                    |                                                                   |
|----------|--------------------|-------------------------------------------------------------------|
| A        | Adjacent category  | Diagnostic explains why past shrinkage happened.                  |
| B        | Inverted operation | Descriptive looks back at what happened, not forward.             |
| C        | Confusion set      | Prescriptive recommends an action; this only estimates shrinkage. |
| <b>D</b> | <b>Correct</b>     | Estimating what will happen next quarter is predictive.           |

## 2. Answer C Concept 1 • transfer

An analysis recommends the order quantity for each store that minimizes expected shrinkage cost. Which type of analytics is this?

**Target word:** analytics type — recommendation **Decides:** It tells managers what to do.

|          |                    |                                                                 |
|----------|--------------------|-----------------------------------------------------------------|
| A        | Adjacent category  | Prediction may feed it, but the output is a recommended action. |
| B        | Confusion set      | Diagnostic explains causes of past results.                     |
| <b>C</b> | <b>Correct</b>     | Recommending an optimal action is prescriptive.                 |
| D        | Inverted operation | Descriptive summarizes the past only.                           |

## 3. Answer B Concept 2

The CFO of a software company ~~with a newly built data lake~~ wants to start by pulling every table into a model to see what patterns appear. Under the IMPACT cycle, what should come first?

**Target word:** first step **Decides:** IMPACT begins with I: identify the question.

|          |                   |                                                                      |
|----------|-------------------|----------------------------------------------------------------------|
| A        | Confusion set     | Testing comes after the question and data are defined.               |
| <b>B</b> | <b>Correct</b>    | A clear question determines which data and tests are needed.         |
| C        | Adjacent category | That is the second step; without a question, profiling has no focus. |
| D        | Camouflage        | More storage does not replace a clear question.                      |

## 4. Answer B Concept 3

An accountant moves 18 months of expense reports from a legacy travel system into the new data warehouse. Dates in the old system are stored as text in three formats, and some employee IDs have leading spaces. ~~The migration budget is \$40,000.~~ In which ETL step should these problems be fixed?

**Target word:** ETL step for cleaning **Decides:** Standardizing formats and trimming values is transformation.

|          |                   |                                                                         |
|----------|-------------------|-------------------------------------------------------------------------|
| A        | Confusion set     | Loading writes already-clean data into the target.                      |
| <b>B</b> | <b>Correct</b>    | Cleaning and standardizing data happens between extraction and loading. |
| C        | Adjacent category | Extraction pulls data as-is from the source.                            |
| D        | Absolute language | Warehouses enforce types but do not reliably repair messy source data.  |



## 5. Answer A Concept 3 • transfer

Clean, standardized data is written into the target data warehouse tables. Which ETL step is this?

**Target word:** name the step — reverse **Decides:** Writing to the target.

|   |                    |                                                          |
|---|--------------------|----------------------------------------------------------|
| A | <b>Correct</b>     | Writing data into the destination is the load step.      |
| B | Inverted operation | Extraction is pulling from the source, the opposite end. |
| C | Confusion set      | Mining analyzes data after it is loaded.                 |
| D | Adjacent category  | Transformation already produced the clean data.          |

## 6. Answer A Concept 4

After extracting the year's 52,318 sales invoices from the ERP, an analyst loads them into a spreadsheet with 52,318 rows, but the dollar total is \$1.2 million less than the general ledger's sales account. ~~The ERP vendor is based in Germany.~~ What should the analyst conclude?

**Target word:** is the extract validated **Decides:** Counts match but the control total does not.

|   |                    |                                                                           |
|---|--------------------|---------------------------------------------------------------------------|
| A | <b>Correct</b>     | Both counts and control totals must agree before the data can be trusted. |
| B | Dropped element    | Ignores the control total that fails.                                     |
| C | Absolute language  | Unexplained differences must be resolved, not assumed away.               |
| D | Inverted operation | Assumes the extract is right without investigating the difference.        |

## 7. Answer A Concept 5

An audit team at a national bank wants to analyze two sources: a table of loan balances with fixed columns, and thousands of scanned loan-officer notes and emails. ~~The bank has 900 branches.~~ Which classification is correct?

**Target word:** classify the data **Decides:** Fixed columns vs. free text and scans.

|   |                    |                                                               |
|---|--------------------|---------------------------------------------------------------|
| A | <b>Correct</b>     | Rows and columns are structured; free text and scans are not. |
| B | Adjacent category  | Confuses data volume with data structure.                     |
| C | Confusion set      | Being digital does not make data structured.                  |
| D | Inverted operation | Reverses the two.                                             |

## 8. Answer C Concept 6

An accounts payable team has three years of invoices, each already labeled by auditors as error or clean. They want a model that labels new invoices the same way. ~~The invoices were scanned by an outside vendor.~~ Which technique fits?

**Target word:** choose the technique **Decides:** Known labels and a categorical target.

|   |                   |                                                                      |
|---|-------------------|----------------------------------------------------------------------|
| A | Dropped element   | Drops the goal of labeling new invoices.                             |
| B | Confusion set     | Clustering ignores labels and finds natural groups.                  |
| C | <b>Correct</b>    | Predicting a known category from labeled examples is classification. |
| D | Adjacent category | Regression predicts a number, not a label.                           |

## 9. Answer C Concept 6 · transfer

An analyst wants to estimate next month's warranty expense in dollars from sales volume. Which technique fits?

**Target word:** choose the technique — numeric target **Decides:** The target is a dollar amount.

|   |                   |                                                                    |
|---|-------------------|--------------------------------------------------------------------|
| A | Absolute language | Some techniques predict numbers rather than groups.                |
| B | Adjacent category | Classification predicts categories, not dollar amounts.            |
| C | <b>Correct</b>    | Predicting a continuous value from another variable is regression. |
| D | Confusion set     | Clustering groups records; it does not predict a value.            |

## 10. Answer B Concept 7

In a company's sales database, the Customers table has a CustomerID field, and the Invoices table also stores CustomerID on each invoice. ~~The database runs on a server in the company's Denver office.~~ What role does CustomerID play in the Invoices table?

**Target word:** key role **Decides:** CustomerID links each invoice to the Customers table.

|   |                    |                                                                           |
|---|--------------------|---------------------------------------------------------------------------|
| A | Confusion set      | A customer has many invoices, so CustomerID cannot identify each invoice. |
| B | <b>Correct</b>     | It references the Customers table's primary key.                          |
| C | Camouflage         | Server location has nothing to do with table keys.                        |
| D | Inverted operation | Uniqueness applies in Customers, not Invoices, where it repeats.          |

## 11. Answer D Concept 7 · transfer

Which field is the best choice for the primary key of an Employees table?

**Target word:** choose a primary key **Decides:** Must be unique and stable for every row.

|   |                   |                                                                |
|---|-------------------|----------------------------------------------------------------|
| A | Dropped element   | Drops the uniqueness requirement; names repeat.                |
| B | Confusion set     | Many employees can start on the same day.                      |
| C | Adjacent category | That is a foreign key to Departments; many employees share it. |
| D | <b>Correct</b>    | Unique, stable and never blank.                                |

## 12. Answer D Concept 8

A controller wants to show the board how monthly operating expenses changed over the past 24 months. ~~The board meets quarterly at the company's new headquarters.~~ Which chart fits best?

**Target word:** chart type **Decides:** Change over time.

|   |                   |                                                                |
|---|-------------------|----------------------------------------------------------------|
| A | Confusion set     | Pies show parts of a whole at one point, not change over time. |
| B | Adjacent category | Scatter plots show relationships between two measures.         |
| C | Absolute language | Well-built charts clarify trends; tables hide them.            |
| D | <b>Correct</b>    | Lines show trends across time periods.                         |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 3             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 0             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 10            |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 6             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 4             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 11            |             |
| Camouflage         | Answers a stem fact that was never the question.             | 2             |             |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                 | Rule                                                                                                                                                       | Trap                                                                       | Tell                                                                  |
|---|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1 | <b>Four types of analytics</b>          | Descriptive = what happened; diagnostic = why; predictive = what will happen; prescriptive = what should we do.                                            | Calling any forecast prescriptive or any dashboard predictive.             | The verb in the request: summarize, explain, estimate, or recommend.  |
| 2 | <b>The IMPACT cycle</b>                 | Identify the question, Master the data, Perform the test plan, Address and refine results, Communicate insights, Track outcomes — in that order.           | Starting by pulling all the data to see what shows up.                     | A project that begins with data rather than a question.               |
| 3 | <b>Extract, transform, load (ETL)</b>   | Extract pulls data from sources; transform cleans, standardizes and restructures it; load writes it into the target store.                                 | Cleaning after loading, or thinking the target system fixes formats.       | Messy formats, spaces, or inconsistent codes during a migration.      |
| 4 | <b>Validating extracted data</b>        | Confirm completeness and accuracy with record counts and control totals reconciled to the source before analysis.                                          | Accepting a data set because row counts match.                             | Counts agree but dollar totals do not.                                |
| 5 | <b>Structured vs. unstructured data</b> | Structured data fits fixed rows and columns; unstructured data (text, images, scans) does not.                                                             | Classifying by size or by being digital.                                   | A table of fields vs. documents, emails or images.                    |
| 6 | <b>Data-mining techniques</b>           | Classification predicts a labeled category; regression predicts a number; clustering groups unlabeled records.                                             | Using clustering when labels exist, or classification for a dollar amount. | Whether the target is a label, a number, or unknown groups.           |
| 7 | <b>Primary and foreign keys</b>         | A primary key uniquely identifies each row in its table; a foreign key is a field that points to another table's primary key.                              | Thinking a linking field must be unique in every table.                    | The same ID appearing in two tables.                                  |
| 8 | <b>Choosing a visualization</b>         | Line charts show change over time; bar charts compare categories; pie charts show parts of a whole; scatter plots show relationships between two measures. | Using a pie for trends or a scatter plot for a time series.                | The phrase over time, by category, share of, or relationship between. |

**Course scope (public description):** Data Analytics for Accountants I introduces basic concepts and various tools and techniques used in the field of data analytics for accounting. Learners will summarize data analysis definitions and models for the accounting field, explore data mining techniques and the extract-transform-load (ETL) process and create a presentation from accounting data results. This course presents a survey of concepts that provides learners with a basic understanding of how data analytics is used in accounting.